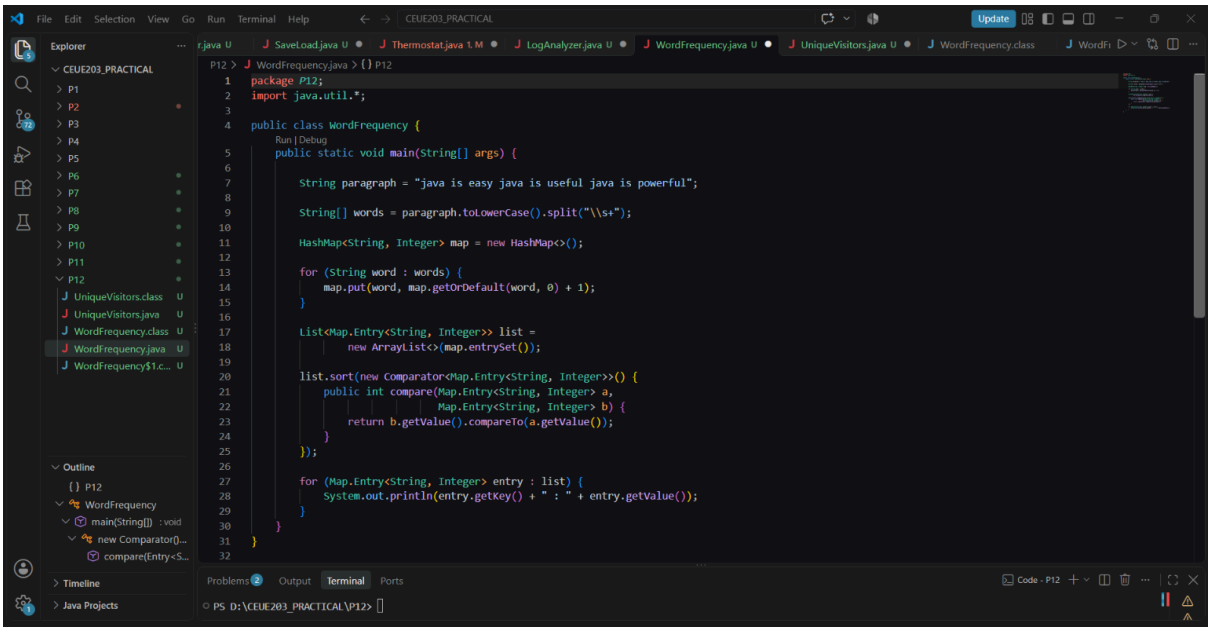
PRACTICAL 12

QUESTION 1

Word frequency: split a paragraph into words, count each in a HashMap, and print the

words sorted by frequency (highest first) using a Comparator.

CODE :



```java
package P12;
import java.util.*;

public class WordFrequency {
    Run | Debug
    public static void main(String[] args) {

        String paragraph = "java is easy java is useful java is powerful";

        String[] words = paragraph.toLowerCase().split("\\s+");

        HashMap<String, Integer> map = new HashMap<>();

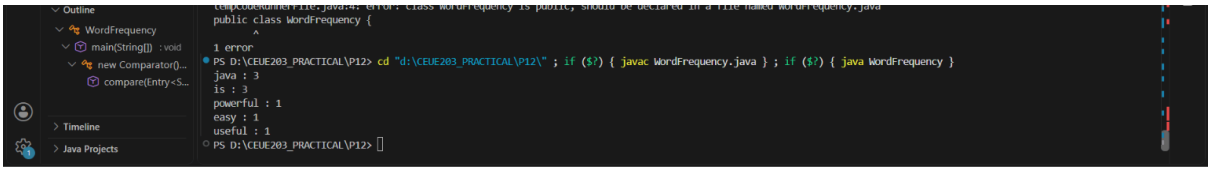
        for (String word : words) {
            map.put(word, map.getOrDefault(word, 0) + 1);
        }

        List<Map.Entry<String, Integer>> list =
                new ArrayList<>(map.entrySet());

        list.sort(new Comparator<Map.Entry<String, Integer>>() {
            public int compare(Map.Entry<String, Integer> a,
                               Map.Entry<String, Integer> b) {
                return b.getValue().compareTo(a.getValue());
            }
        });

        for (Map.Entry<String, Integer> entry : list) {
            System.out.println(entry.getKey() + " : " + entry.getValue());
        }
    }
}
```

OUTPUT :



```
public class WordFrequency {
       ^
1 error
PS D:\CEUE203_PRACTICAL\P12> cd "d:\CEUE203_PRACTICAL\P12\" ; if ($?) { javac WordFrequency.java } ; if ($?) { java WordFrequency }
java : 3
is : 3
powerful : 1
easy : 1
useful : 1
PS D:\CEUE203_PRACTICAL\P12>
```

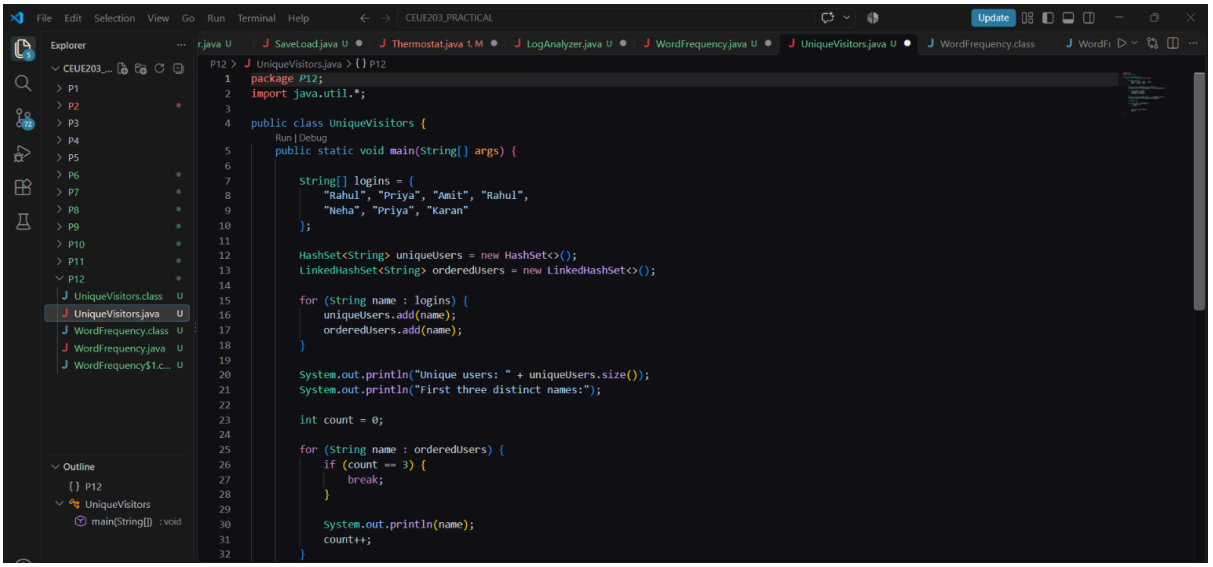
QUESTION 2

Unique visitors: from a list of login names with repeats, use a HashSet to count unique

users and a LinkedHashSet to keep first-seen order; print the unique count and the first

three distinct names.

CODE :



```java
package P12;
import java.util.*;

public class UniqueVisitors {
    Run | Debug
    public static void main(String[] args) {

        String[] logins = {
            "Rahul", "Priya", "Amit", "Rahul",
            "Neha", "Priya", "Karan"
        };

        HashSet<String> uniqueUsers = new HashSet<>();
        LinkedHashSet<String> orderedUsers = new LinkedHashSet<>();

        for (String name : logins) {
            uniqueUsers.add(name);
            orderedUsers.add(name);
        }

        System.out.println("Unique users: " + uniqueUsers.size());
        System.out.println("First three distinct names:");

        int count = 0;

        for (String name : orderedUsers) {
            if (count == 3) {
                break;
            }

            System.out.println(name);
            count++;
        }
```

OUTPUT :



```
java : 3
is : 3
powerful : 1
easy : 1
useful : 1
PS D:\CEUE203_PRACTICAL\P12> cd "d:\CEUE203_PRACTICAL\P12\" ; if ($?) { javac UniqueVisitors.java } ; if ($?) { java UniqueVisitors }
Error: Could not find or load main class UniqueVisitors
Caused by: java.lang.NoClassDefFoundError: UniqueVisitors (wrong name: P12/UniqueVisitors)
PS D:\CEUE203_PRACTICAL\P12> cd "d:\CEUE203_PRACTICAL\P12\" ; if ($?) { javac UniqueVisitors.java } ; if ($?) { java UniqueVisitors }
Unique users: 5
First three distinct names:
Rahul
Priya
Amit
PS D:\CEUE203_PRACTICAL\P12>
```

KEY QUESTIONS

3. What is the difference between Comparable and Comparator?

1. **What is the difference between a List, a Set and a Map?**

- **List:** Stores elements in order and allows duplicate values. Example: ArrayList.
- **Set:** Stores unique elements and does not allow duplicates. Example: HashSet.
- **Map:** Stores data in key-value pairs, where each key is unique. Example: HashMap.

2. **When do you use a HashMap versus a TreeMap?**

- **HashMap:** Use it when you need fast insertion and searching without sorting the keys.
- **TreeMap:** Use it when you need keys to be stored in sorted order.

3. **What is the difference between Comparable and Comparator?**

- **Comparable:** Defines the natural ordering of objects using the compareTo() method inside the class.

- **Comparator:** Defines a custom ordering of objects using the compare() method, usually in a separate class or with a lambda expression.